Installing ROOT on Windows

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*Minimal changes and additions for the March 2026 I2TheTerrascale by A.P., a fellow student at NKUA

1 Introduction

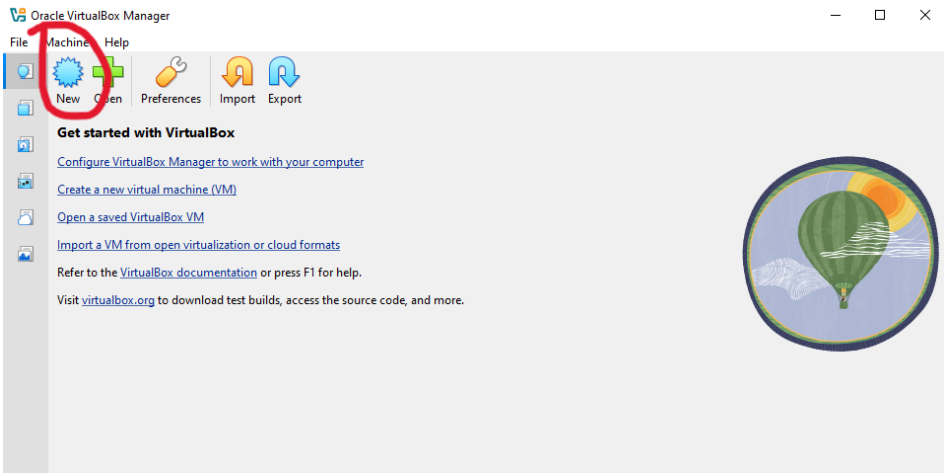
The ROOT framework works best in a Linux or Mac OS and difficulties arise when one tries to install it on Windows. A way to overcome this without actually making the switch to a different OS, is by creating a "smaller computer", inside of your own, through the use of a virtual machine!

2 Necessary Installations

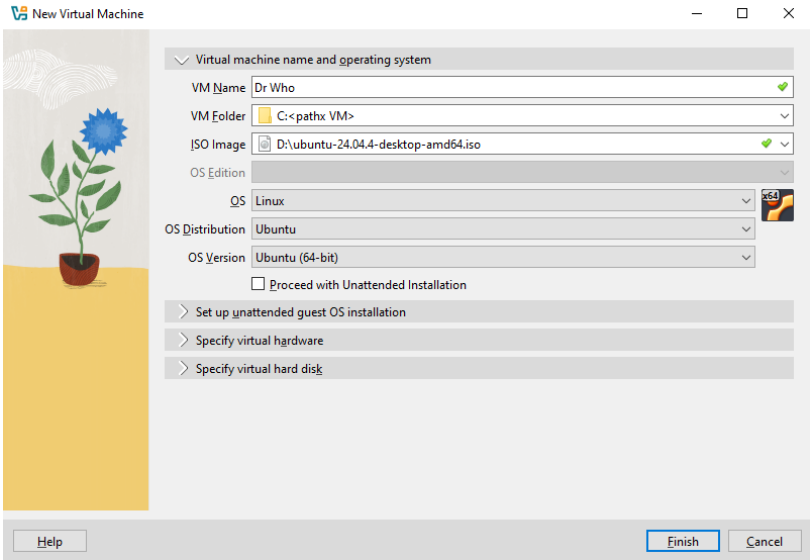
- Initially you will need to install Virtual Box, you can do this by simply pressing "Windows Hosts" here.
- For your next step, you will download the Ubuntu Linux distribution here. ~~Ubuntu 24.04.2~~ Ubuntu 24.04.4 has been used for the purposes of this guide.

3 Creating the Virtual Machine

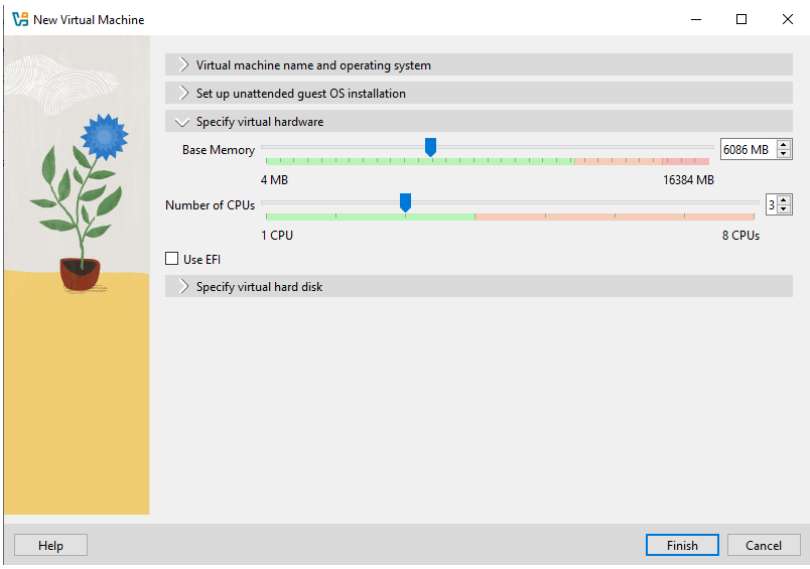
After the installations, you are ready to set up the Virtual Machine. Open up Virtual Box, and press the button that says "New"



After pressing this, name your Virtual Machine however you like and pick the file where you want to store it inside your computer. In the "ISO Image", please choose the .iso file that corresponds to the Ubuntu version you downloaded in the previous step.

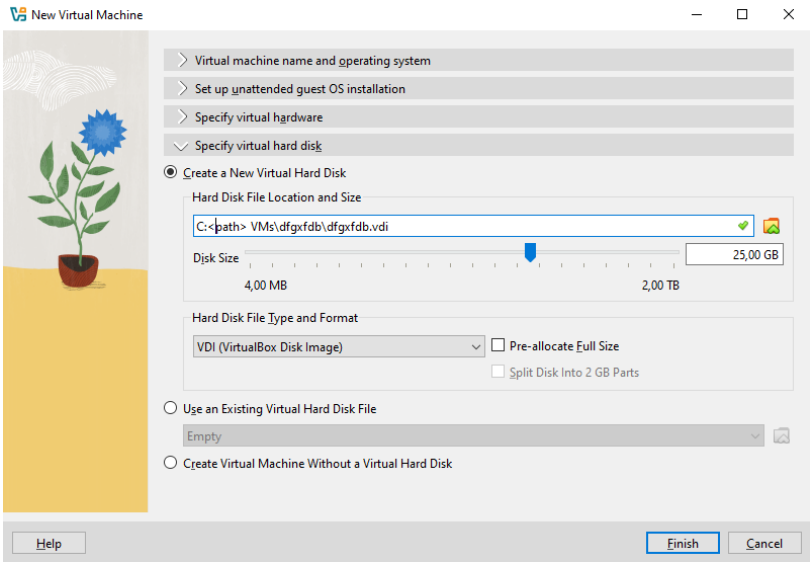


Important!!!: DO NOT select Procced with Unattended Installation, this part is **crucial**. This needs to be so, for us to be able to install Linux manually and us to be able to access the terminal.



For this step we choose the amount of RAM memory and number of processors we want to allocate for the function of the Virtual Machine. This part seems innocent, but is quite tricky and what one chooses depends on their computer. Generally, you will want to allocate enough RAM for your installation and general tasks inside the Virtual Machine to run smoothly, but if you allocate more memory than what is generally available in your computer, the Virtual Box app will **not** work properly. My recommendation is to use a minimum of 4 GB of RAM and 2 processors and a maximum of 6-8 GB and 4-5 processors depending on the specifications of your PC/laptop.

Something important regarding this part, is that some processors have an option turned off regarding *virtualization*, this basically disallows us to use more than one processor for our virtual machine. Should we try to use more, Virtual Box will make sure to tell us, f.e. for AMD processors the error **AMD-V is disabled in the BIOS** will appear. This part can be easily fixed, but one would have to access the BIOS of their computer and alter the CPU configurations in order to allow more than one processor to be used in a Virtual Machine. Since accessing the BIOS and navigating through it, is personalized in every computer and is related to the motherboard, should you encounter this problem, use Youtube and/or any AI to guide you through the changes that are needed.



Finally you will need some free space in your hard drive for the Virtual Machine to create it's own partition which will be used for Linux to be able to operate. Using about 25-30 GB of disk space will be sufficient for our purposes.

4 Installing Linux

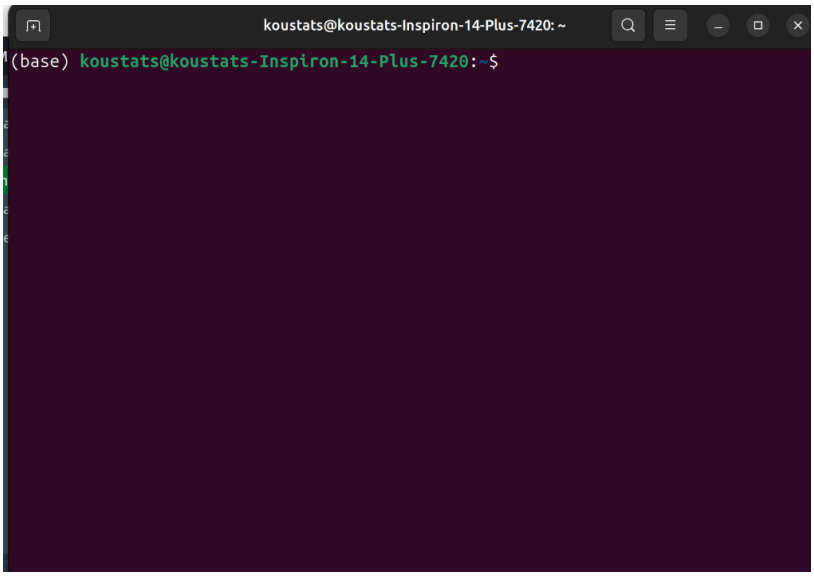
After you have set up the Virtual Machine, you are ready to start it and go through the manual installation of Ubuntu. If you are unfamiliar with this process, you should not worry, since the manual installation of Ubuntu is quite user-beginner friendly, but nonetheless a useful youtube video that guides you through the process can be found here (you might see some differences regarding the Virtual Box setup, don't worry too much about them, the later part of the video which focuses on the installation of Ubuntu is what you want to focus on if you need help with that).

5 Using Linux

You are now ready to start! Our biggest friend for installations and doing work on ROOT later on will be the Terminal. Go ahead and press the Ubuntu logo on the lower left part of your screen and look up "Terminal". I would recommend to right click it and pin it in your taskbar (which should be located in the left part of your screen), because you will be needing it quite a lot. It would also be helpful if you could access the guide inside the Virtual Machine (perhaps use Firefox to access your email and send this file to yourself) or you can follow the instructions below to enable it Window-VM Linux cross copy-paste.

6 Using a terminal

Go ahead and open the Terminal, you should see something like this:



If you have not used a Terminal before (also called Command Prompt on Windows), this is not a problem at all, most of what you will be doing is copying and pasting commands that I will provide you in this guide (to paste to the terminal in Linux, you need to use Ctrl+Shift+V. In any other app within Linux, the paste key combination is the classic Ctrl+V).

7 Copy-Paste across Windows and Linux (Optional)

If you want to be able to cross copy-paste from Windows to Linux and the reverse, you can follow the following steps. In the VM menu bar at the top of the window click:

Devices → Insert Guest Additions CD image

A disk should appear inside the vertical Ubuntu bar menu. In the terminal run:

```
sudo apt install build-essential dkms linux-headers-$(uname -r)
```

Then mount and run the installer:

```
cd /media/$USER/VBox_GAs_*
sudo sh VBoxLinuxAdditions.run
```

Having done that, reboot:

```
sudo reboot
```

Finally, go again to the VM window menu and select:

Devices → Shared Clipboard → Bidirectional

8 Prerequisites

In this part we will do the necessary installations for us to be able to download and use ROOT. Open the Terminal, where we are going to run some commands.

The first ones we need to run are:

```
sudo apt update
sudo apt upgrade
```

Those will just update Linux. The next commands we will run are:

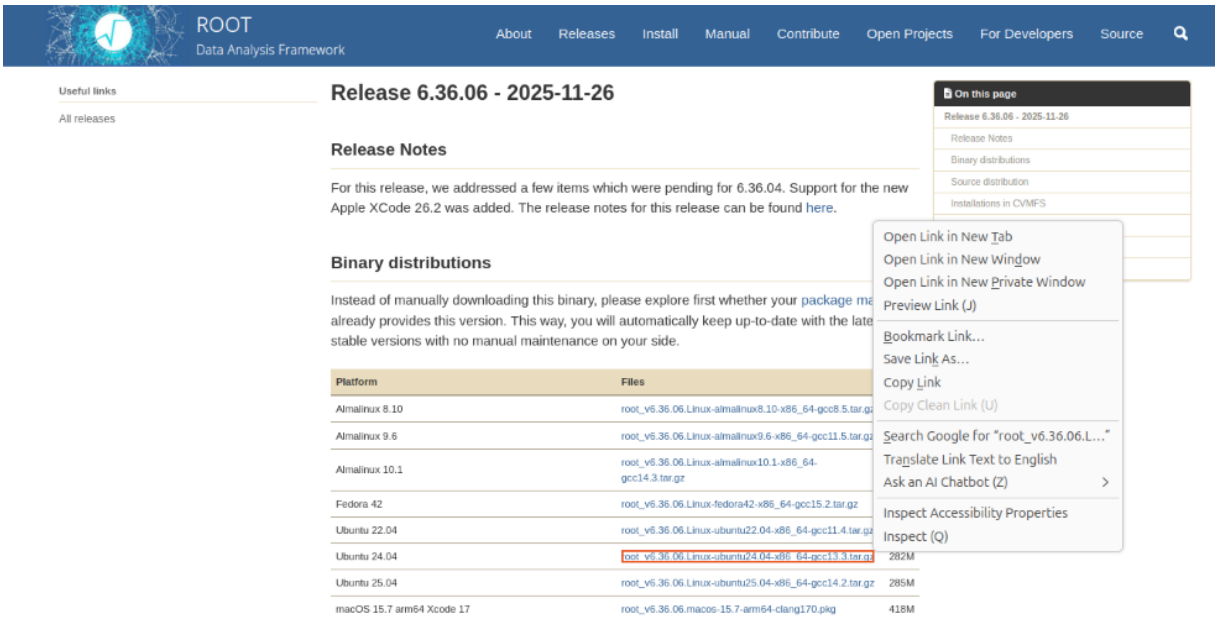
```
sudo apt-get update
sudo apt install binutils cmake dpkg-dev g++ gcc libssl-dev git libx11-dev \
libxext-dev libxft-dev libxpm-dev python3 libtbb-dev libvdt-dev libgif-dev
```

Please note that after every time you run a command that installs something on your computer, you will be asked to provide the password you chose when you installed Ubuntu, so don't forget that! A lot of gibberish will probably show up on your terminal, some times you will be asked to confirm that you want an installation (you will be able to do this by typing 'y' and Enter on your terminal), but for now you will just have to be patient and wait for all the downloads to finish. If you encounter any problems during this installation that are not part of this guide, take a screenshot of the error messages and give them to an AI (GPT, Deepseek etc.) or look it up on Youtube and you should be able to resolve it.

P.S. If your terminal feels too "crowded" you can clear it by typing *clear*, and if you somehow mess up a command, you can use the *up* arrow to recover a command that you previously ran!

9 ROOT! Finally!

I hope that you are not tired, because after the steps you followed, you are now ready to download ROOT and run your first program! Initially you want to visit this site, and then go to Releases and click the latest stable release.



Your screen should look like this. Now go ahead and find the version of ROOT that corresponds to the version of Linux you downloaded and are using in this virtual machine (in this guide I use ~~Ubuntu 22.04~~ Ubuntu 24.04 and root version 36.06). Instead of pressing the link and downloading the version through your browser, copy the link as you can see in the screenshot. Now go to your terminal and paste the command (which will probably require modification):
Please note that the following two commands might not work for you, you need to use the appropriate name of the file that corresponds to the version of ROOT you want to download!

```
wget https://root.cern/download/root_v6.36.06.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz
```

Now to compile what you just downloaded, run the command (after you have modified it accordingly, if needed):

```
tar -xzvf root_v6.36.06.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz
```

And now ROOT should be installed! Now, for us to be able to use ROOT after we open the Terminal we basically have to tell ROOT "where to find us", this can be done manually, but we can automate it! For a start write in your terminal:

```
sudo apt install gedit
gedit ~/.bashrc
```

Then go to the end of the file and paste the following command:

```
source root/bin/thisroot.sh
```

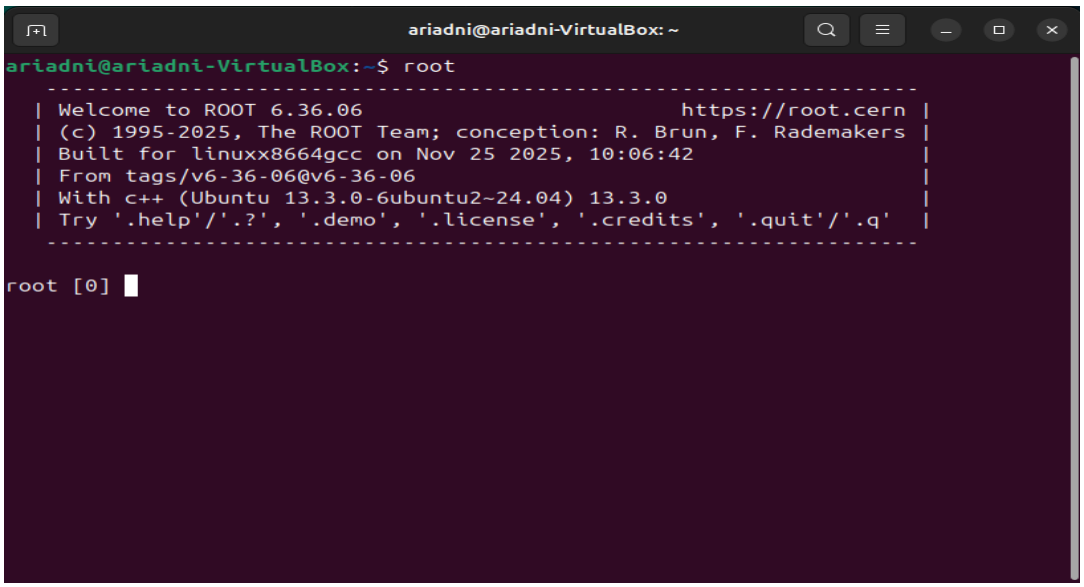
Save your changes and close the editor. Finally run the command:

```
source ~/.bashrc
```

In a nutshell the script bashrc runs each and everytime you open a terminal, in this way we help "ROOT find us" every time we open the terminal automatically.

10 Does everything work?

To ensure everything works correctly, type root in the terminal. The following should appear:



Now stop ROOT by pressing Ctrl+Z or by typing .q .

Before running the code needed for the preschool exercise, we need to download the davix package, required to download information over the Internet:

```
sudo apt update
sudo apt install davix
```

Now, try running your first ROOT program by pasting this in the terminal:

```
wget https://github.com/theofil/I2TheTerascale/archive/refs/heads/main.zip
unzip main.zip
cd I2TheTerascale-main/code/C/
root -l makePlot.C
```

It should be easy to understand by the output if everything has went according to plan :)

11 Credits

This guide's predecessor can be found here and was made by Xenophon Karyophyllis, another student at NKUA.